Lab/Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		**25%**	**50%**	**75%**	**100%**	**25**
Problem understanding & Analysis	**7**					
Implementation	**8**					
Report writing	**10**					
					Total obtained mark	
Comments						

Semester: Fall …2026……

Team name: Code Ninjas

Aranna Saha || **241-35-197**

Abdullah-Al-Arafat Sirat || **241-35-057**

Sabbir Hasan Emon || **241-35-012**

Batch: 42 **Section: B1**

Course Code: SE-331 **Course Name: Software Engineering Design And Capstone Project**

Course Teacher Name: Rahat Uddin Azad
Designation: Lecturer
Submission Date: 3-10-2026

Project Proposal

AgriSense: A Bangla-First AI Mobile Platform for Crop Disease Detection, Crop Advice and a Verified Agri Marketplace

1. Introduction

AgriSense is a Bangla-first mobile platform for marginal farmers in Bangladesh. A farmer can take a photo of a sick leaf and get a diagnosis, ask a question by voice instead of typing, and receive crop suggestions based on soil, district and weather. The recommended seeds, fertilizers and medicines can then be bought from licensed vendors through a verified marketplace called Haat_Bazar. The idea is kept focused on a few crops and a few features, with AI at the center of every main workflow.

1.1 Background Overview

The application domain is agriculture. Farming supports a very large number of families in Bangladesh, and most of them work on small plots. When something goes wrong in the field, a farmer usually asks a neighbor, an elder, or the local fertilizer and pesticide shop. Government extension officers (the SAAOs under DAE) do give advice, but one officer is responsible for many villages, so a farmer cannot always reach one at the moment a problem appears.

Some digital tools already exist. Plantix can diagnose plant problems from photos, Fosli by ACI offers weather and advisory information, and Krishoker Janala from DAE and a2i is a database of crop symptoms and solutions. Even so, from what we reviewed, AI is used very little by marginal farmers in practice. Most of these tools expect the user to read a lot of text, type queries or browse through menus, and none of them connects the diagnosis to the medicine that is actually sold nearby by a verified seller.

1.2 Problem Statement

A farmer who sees spots or curling on a crop has to guess the cause. The usual way is to describe the problem to a shopkeeper, who then suggests a product. If the guess is wrong, the farmer spends money on the wrong spray, loses time, and the disease keeps spreading. Good advice does exist in DAE and BARC publications, but it is written in technical language and is not available at the moment of need.

Gaps in existing systems

- Interfaces are text-heavy and need typing, which is hard for farmers with low literacy.
- Diagnosis results do not lead to a purchase. The farmer still has to find the right product and a trustworthy seller alone.
- Crop advice is often general and not tied to the farmer's own district, soil type and current weather.
- Counterfeit or low-quality inputs are a known risk in local markets, and there is no easy way for a farmer to check a seller.

Who is affected

Marginal and small farmers are affected the most. Vendors who sell genuine inputs are affected too, because they have no easy way to reach farmers. Extension officers are overloaded with routine questions that could be answered faster by a system.

Why AI is needed

Today the identification of a plant disease is done manually, by a person looking at the symptoms and comparing them with what they remember. Questions are answered by people reading or recalling guidelines. Choosing a crop for the season is done by experience or by reading bulletins. AI can take over these specific jobs:

- Image classification can recognize a disease from a leaf photo in seconds, which replaces the visual guesswork.
- Bangla speech recognition lets a farmer speak instead of type.
- Retrieval-based answering over verified documents lets the system give advice from DAE and BARC material in simple Bangla.
- A trained recommendation model replaces manual reading of soil and season tables.

We are keeping the problem realistic. The first version will only cover a few major crops and a limited list of common diseases. It will support the work of agricultural experts and will never claim to replace them.

1.3 Objectives

General Objective: To build a working prototype of a Bangla-first mobile app that helps small farmers identify diseases in selected crops, get suitable advice by voice or text, and buy the recommended inputs from verified vendors.

AI Objective: To build and connect the following AI components so that the main workflow of the app depends on them:

- A crop disease classifier for rice, potato and tomato (about 12 disease and healthy classes) that targets at least 85% accuracy on a held-out test set and shows a confidence score with every result.
- A Bangla voice assistant that turns a spoken question into text, finds the answer in a verified knowledge base, and reads it back in Bangla.
- A crop recommender that suggests the top three suitable crops from district, soil type, season and current weather.
- A product matching step that links each diagnosis to the treatment products listed by verified vendors.

The 85% figure is our target and not a promise. We will report the real numbers from testing, including cases where the model does badly.

1.4 Scope

In scope

- Mobile app (Android) with Bangla and English interface, large buttons, icons and a high-contrast mode.
- Registration and login for farmers, vendors and experts, plus an admin panel.
- Photo-based disease detection for rice, potato and tomato, with treatment advice and confidence score.
- Bangla voice questions and spoken answers.
- Soil, district and weather based crop suggestions, and a simple seasonal calendar.
- Haat_Bazar: product listing, search, stock update, order request, order status tracking and a digital invoice.
- Expert review of low-confidence diagnoses and a way for experts to update the knowledge base.
- Weather alerts through an external weather API.
- Basic offline support: view saved history and queue a photo to upload when the connection returns.

Out of scope

- IoT sensors, drones or any special hardware.
- Online payment gateways. Orders in the prototype will use cash on delivery or payment between buyer and seller.
- Delivery logistics and courier integration.
- Insect pest identification and crops other than the three selected ones.
- An iOS release and Play Store publishing. These are future work.
- On-device AI inference. The first version runs the models on the server.

What the AI will and will not do

The AI will classify a leaf photo, estimate its own confidence, convert Bangla speech to text and back, retrieve and summarize advice from verified documents, and rank crops for a given district and season. It will not give a final or guaranteed diagnosis, it will not recommend pesticide doses that are not found in the verified knowledge base, and it will not answer questions outside agriculture. If confidence is low, the app sends the case to a human expert.

System boundaries

Only the features listed above are part of this project. Any new idea during development goes into the Future Work list unless the whole team and the course teacher agree to add it.

1.5 Stakeholders

Stakeholder	Role	Interest in the system
Farmers	Main users. Take photos, ask questions, view advice and place orders.	Correct diagnosis, simple language, lower crop loss and cost.
Vendors	Licensed sellers who list seeds, fertilizers and medicines and handle orders.	A wider customer base and a trusted seller profile.
Agri Experts (SAAOs, specialists)	Review low-confidence cases and keep the knowledge base up to date.	Less time on routine questions and a say in what advice the system gives.
System Administrator	Verifies vendors, manages users and monitors the system.	A safe marketplace and a system that runs smoothly.
Development team	Design, build, test and document the system.	A working and well-tested capstone project.
Course teacher / university	Supervise and evaluate the project.	A project that is realistic, complete and original.
Organizations such as DAE, SRDI and BARC	Sources of agricultural data and guidelines. Not directly involved in the prototype.	Their information should be used correctly and not misrepresented.

1.6 Proposed Solution

AgriSense connects the steps a farmer actually goes through: spotting a problem, finding out what it is, learning what to do, and getting the right product. The workflow is as follows.

- **Step 1, Snap & Heal.** The farmer opens the camera and photographs the affected leaf. The photo goes to the server where the disease classifier returns the disease name and a confidence score. (AI)
- **Step 2, Advice.** The system looks up the treatment steps for that disease in the verified knowledge base and shows them in Bangla, with a spoken version. (AI: retrieval and speech)
- **Step 3, Expert check.** If confidence is below a set threshold, the case is sent to an expert who confirms or corrects it.
- **Step 4, Shovon-Boli.** At any point the farmer can ask a follow-up question by voice, such as when to spray or how much to use. (AI: speech recognition and RAG)
- **Step 5, Soil-Soul Match.** Before a new season the farmer selects the district and soil type, and the recommender suggests suitable crops using the weather forecast. (AI)
- **Step 6, Haat_Bazar.** The recommended products from the diagnosis appear as a list from verified vendors with price and stock. The farmer can place an order and follow its status.

The diagram below gives a simple overview. The orange blocks are the AI components.

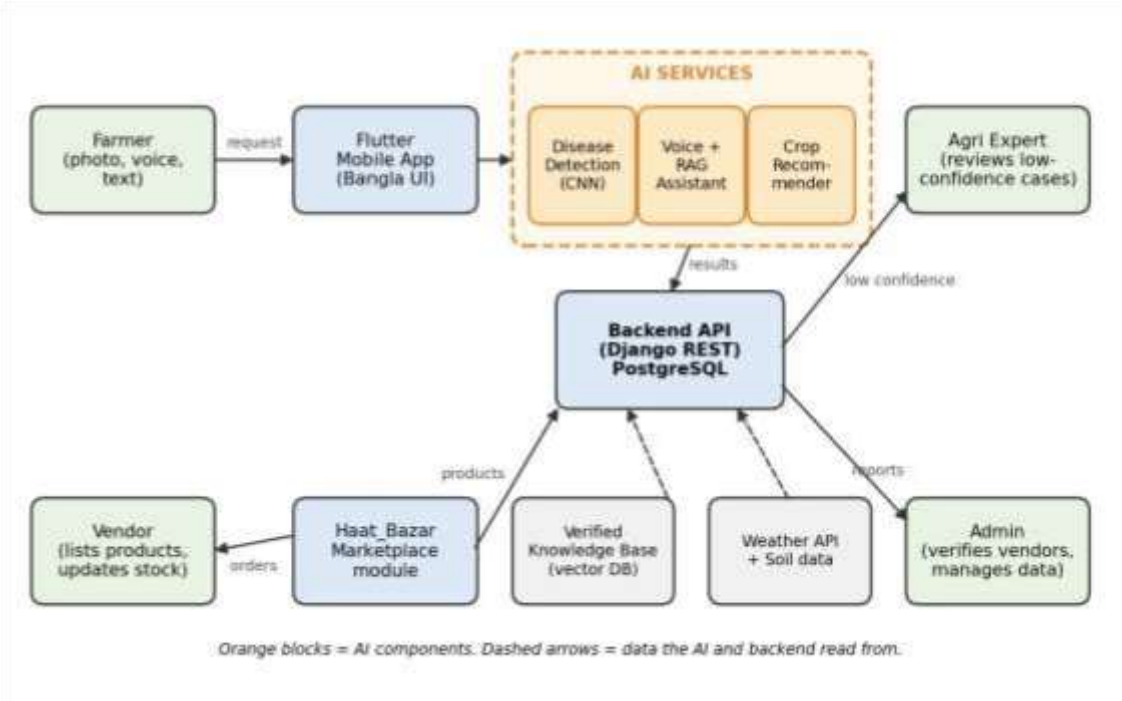


Figure 1: AgriSense system overview with AI components marked

1.7 AI Features

Existing AI Features

Existing system	AI or smart features available	Limitations we found
Plantix	Image-based diagnosis of plant diseases and pests.	Not connected to verified local vendors. Voice interaction is limited, and the advice is general rather than based on a farmer's district.
Krishoker Janala (DAE & a2i)	A visual database of symptoms with image-based identification.	The farmer has to search and read through content. No voice interface and no purchase step.
Fosli (ACI)	Weather and advisory information.	Advice is mostly informational. It does not diagnose from a photo and does not give a personal crop plan.

These notes come from our own review of the apps and their public descriptions, so some details may have changed.

Proposed AI Features

Feature	Input	Output	Manual work it replaces
Snap & Heal (disease detection)	Photo of a crop leaf.	Disease name, confidence score, and treatment steps.	Asking neighbors or shopkeepers to guess the disease.
Shovon-Boli (voice assistant)	Bangla spoken question.	Short answer in Bangla text and voice, taken from verified documents.	Typing queries, or traveling to find an extension officer for simple questions.

Feature	Input	Output	Manual work it replaces
Soil-Soul Match (crop recommender)	District, soil type, season, weather data.	Top three suitable crops with a short reason.	Reading soil and season tables, or relying on habit.
Treatment-toproduct matching	Diagnosis and treatment text.	A ranked list of matching products from verified vendors.	Searching for the right product and seller by oneself.

AI Techniques

- **Computer vision:** a convolutional neural network (MobileNetV2 or EfficientNet-Lite using transfer learning) trained on public plant disease datasets plus images we collect from local fields.
- **Speech AI and Bangla language AI:** a Bangla speech-to-text and text-to-speech service for voice input and output.
- **RAG with an LLM:** verified DAE, BARC and SRDI documents are split into chunks, stored in a vector database, and retrieved to ground the LLM answer so it does not invent advice.
- **Recommendation:** a decision tree or random forest model trained on district-level soil and crop suitability data, combined with simple expert rules.
- **Semantic search:** embeddings are used to match a treatment to vendor products, even when the names are written differently.

Why AI Is Essential

If all the AI features were removed, AgriSense would be left with a login screen and a list of products, which is an ordinary shopping app. There would be no diagnosis, no voice help and no crop suggestions, and the marketplace would have no reason to be linked to the farmer's problem. In our design every main path starts with an AI output: the photo becomes a diagnosis, the diagnosis becomes a product list, and the voice question becomes advice. The marketplace is the final step of that chain and not a separate feature added beside it.

2. System Requirements

The main functions of the system are listed below. Each statement is written so that it can be tested.

Functional requirements

ID	Requirement
FR1	The system shall allow farmers, vendors and experts to register with name, mobile number, role and location.

FR2	The system shall let users log in securely and shall send them to the dashboard for their role.
FR3	The system shall let a user reset a password using an SMS OTP.
FR4	The system shall let users edit their profile, farm location and crop preferences.
FR5	The system shall let a farmer capture a leaf photo with the in-app camera or upload one from the gallery.
FR6	The system shall identify the disease from an uploaded leaf photo of rice, potato or tomato and show a confidence score. (AI)
FR7	The system shall show treatment steps in Bangla for each detected disease, taken from the verified knowledge base. (AI)
FR8	The system shall ask the farmer to retake the photo, or offer expert review, when confidence is below the set threshold. (AI)
FR9	The system shall save each diagnosis in the farmer's history, which can be viewed later.
FR10	The system shall accept a Bangla voice question and convert it to text. (AI)
FR11	The system shall answer the question using retrieved content from the verified knowledge base and say so when no answer is found. (AI)
FR12	The system shall read answers aloud in Bangla. (AI)
FR13	The system shall recommend the top three crops for the selected district, soil type and season. (AI)
FR14	The system shall fetch weather data for the farmer's district and send alerts for heavy rain, heat or similar risks.
FR15	The system shall show a seasonal crop calendar for the selected crop.
FR16	The system shall suggest matching products from verified vendors after a diagnosis. (AI)
FR17	The system shall let farmers search and filter products by name, category, price and availability.
FR18	The system shall let vendors add, edit and delete products and update stock.
FR19	The system shall let farmers place an order, track its status (Pending, Shipped, Delivered) and receive a digital invoice.
FR20	The system shall let an admin verify or suspend vendors and manage user accounts.
FR21	The system shall let experts review flagged diagnoses and update the disease and treatment knowledge base.
FR22	The system shall let users submit ratings and feedback on advice and on vendors.
FR23	The system shall store photos taken offline and upload them when internet is available.

Non-functional requirements

Category	Requirement
Performance	On a normal 4G connection, the disease result should appear within about 5 seconds of upload.
Accuracy	The disease classifier shall reach at least 85% accuracy on the test set, and test results shall be reported honestly.
Usability	The interface shall be icon-based with large buttons, and a farmer should be able to finish a diagnosis in no more than four taps after opening the app.
Accessibility	The app shall have a high-contrast mode for outdoor use.
Localization	All screens and advice shall be available in Bangla.
Security	Passwords shall be hashed, communication shall use HTTPS, and photos and personal data shall be stored securely.
Reliability	The system should stay available during pilot testing, with problems logged so they can be fixed.
Maintainability	The code shall be modular, version-controlled on Git and documented.

3. Tools and Technologies

Area	Tools
Programming languages	Dart (mobile app), Python (backend and AI), SQL.
Mobile framework	Flutter, with screen designs prepared in Figma before development.
Backend	Django with Django REST Framework, JWT for authentication, OTP through an SMS gateway.
Database and storage	PostgreSQL for application data, with cloud storage for images. A vector database (ChromaDB or pgvector) for the knowledge base.
ML framework and libraries	TensorFlow/Keras, OpenCV, NumPy, Pandas, scikit-learn. Training on Google Colab.
AI models and APIs	MobileNetV2 or EfficientNet-Lite for disease classification. A Bangla speech-to-text and text-to-speech API. An LLM API for answer generation in the RAG pipeline. A multilingual embedding model for retrieval and semantic search.
Datasets and knowledge sources	Public plant disease datasets (such as PlantVillage and rice leaf disease sets from Kaggle), images we collect locally, and verified documents from DAE, BARC and SRDI.
External services	A weather API (such as OpenWeatherMap) for forecasts and alerts.
Hosting	A cloud platform (AWS or GCP student credits) for the backend and models.
Design and testing	Figma, Postman, Git and GitHub, and Flutter and Django testing tools.

4. Project Timeline and Work Plan

The project runs for 16 weeks. We use an iterative approach, so each phase ends with something that can be demonstrated.

Milestones

Weeks	Milestone	Main tasks and output	Lead
1 to 3	Requirements and data	Finalize requirements. Collect and label crop images. Gather soil and district data and verified documents. Meet an agri expert to confirm the disease list.	All
4 to 7	AI development	Train and evaluate the disease classifier. Build the crop recommender. Set up the RAG pipeline and knowledge base. Output: working models with test results.	Aranna
8 to 10	App development	Build Flutter screens in Bangla. Integrate camera, voice input and output, and the marketplace screens.	Sirat
11 to 12	Backend and integration	Build APIs, authentication, database and marketplace logic. Connect the AI services to the app. Add the offline photo queue.	Emon
13 to 14	Testing	Unit, integration and AI testing. Fix bugs and measure the model on unseen images.	All
15 to 16	Pilot and report	Trial with a small group of farmers and an SAAO. Collect feedback, make final fixes, finish the report and prepare the demo.	All

Task breakdown among team members

Team member	Main responsibilities	Secondary role
Abdullah-Al-Arafat Sirat (241-35-057)	Flutter app and Bangla UI. Voice input and output screens. Marketplace screens. Highcontrast mode and offline queue.	Usability testing and farmer feedback.
Aranna Saha (241-35-197)	Collecting and cleaning images. Training and testing the disease classifier. Crop recommender. RAG pipeline and knowledge base.	Helping with model deployment on the backend.
Sabbir Hasan Emon (241-35-012)	Project planning and coordination. Django backend and REST APIs. Database. Authentication and security. Integrating AI services.	Documentation and final report.

All three members share system testing, field testing with farmers and SAAOs, validation with agricultural experts, and presentation preparation.

5. Optional Sections

5.1 Existing System

The existing apps were discussed in section 1.7. In short, Plantix is strong at image diagnosis, Krishoker Janala holds local symptom knowledge, and Fosli offers weather and advisory content. AgriSense tries to bring the useful parts together in Bangla, add voice interaction, and connect the result to a verified marketplace.

5.2 System Architecture

The system has four layers:

- **Client layer:** the Flutter mobile app, and a simple web panel for admins and experts.
- **Application layer:** a Django REST backend that handles authentication, orders, products, history and the connection to the AI services.
- **AI layer:** the disease classifier, the speech and RAG assistant, and the crop recommender, each exposed as a service the backend can call.
- **Data layer:** PostgreSQL, image storage, the vector database of verified documents, and the weather API.

5.3 Ethical, Legal and Social Considerations

- **Data privacy:** we will ask for consent before collecting photos, location and phone numbers, encrypt stored data, and collect only what the app needs.
- **AI bias:** the training images will come mostly from public datasets, which may not match Bangladeshi fields. We will add locally collected images, test on them separately and report any weak classes.
- **Wrong advice:** a wrong diagnosis can cost a farmer money. The app will always show the confidence score, send low-confidence cases to an expert and include a note that the result is guidance and not a guarantee.
- **Chemical safety:** advice on pesticides will come only from verified documents and will include safety instructions.
- **Vendor trust:** only vendors approved by the admin can list products, and farmers can rate them.
- **Inclusion:** voice interaction, large icons and Bangla language are meant to help people who are often left out by text-based apps.

5.4 Future Work

- Run the disease model on the phone (TensorFlow Lite) so it works without internet.
- Add more crops, and insect pests as well as diseases.
- Add mobile payment options such as bKash and Nagad.
- Release an iOS version and publish on the Play Store.
- Connect with DAE or NGO platforms for larger scale use and richer data.